The Asset Pipeline

1 What is the Asset Pipeline?

The asset pipeline provides a framework to concatenate and minify or compress JavaScript and CSS assets. It also adds the ability to write these assets in other languages and pre-processors such as CoffeeScript, Sass and ERB.

The asset pipeline is technically no longer a core feature of Rails 4, it has been extracted out of the framework into the sprockets-rails gem.
The asset pipeline is enabled by default.

You can disable the asset pipeline while creating a new application by passing the `--skip-sprockets` option.

```
rails new appname --skip-sprockets
```

Rails 4 automatically adds the `sass-rails`, `coffee-rails` and `uglifier` gems to your Gemfile, which are used by Sprockets for asset compression:

```
gem 'sass-rails'
gem 'uglifier'
gem 'coffee-rails'
```

Using the `--skip-sprockets` option will prevent Rails 4 from adding `sass-rails` and `uglifier` to Gemfile, so if you later want to enable the asset pipeline you will have to add those gems to your Gemfile. Also, creating an application with the `--skip-sprockets` option will generate a slightly different `config/application.rb` file, with a require statement for the sprockets railtie that is commented-out. You will have to remove the comment operator on that line to later enable the asset pipeline:

```
# require "sprockets/railtie"
```

To set asset compression methods, set the appropriate configuration options in `production.rb` - `config.assets.css_compressor` for your CSS and `config.assets.js_compressor` for your JavaScript:

```
config.assets.css_compressor = :yui
config.assets.js_compressor = :uglifier
```

The `sass-rails` gem is automatically used for CSS compression if included in Gemfile and no `config.assets.css_compressor` option is set.

1.1 Main Features

The first feature of the pipeline is to concatenate assets, which can reduce the number of requests that a browser makes to render a web page. Web browsers are limited in the number of requests that they can make in parallel, so fewer requests can mean faster loading for your application.

Sprockets concatenates all JavaScript files into one master `.js` file and all CSS files into one master `.css` file. As you'll learn later in this guide, you can customize this strategy to group files any way you like. In production, Rails inserts an MD5 fingerprint into each filename so that the file is cached by the web browser. You can invalidate the cache by altering this fingerprint, which happens automatically whenever you change the file contents.